Essential in CATIA (40 Hours)				
S.No	**Session**	**Topics**	**Type**	**Duration (Hrs)**
1	Session 1	**Introduction to CATIA V5** What is CATIA V5 Design intent System requirements Getting started **Understanding CATIA interface** Workbenches Menus and toolbars Finding tools Specification tree Compass Graphic properties Message bar Document management **Introduction to sketcher** Entering sketcher workbench Basic sketching Positioned sketching **Profile creation** Profile Predefined profile Circle Spline Conic Line Point Axis **Understanding standard toolbar** New Open Save	Theory/Demo	2.0 Hrs
2	Session 2	Exercises included for practice	Practicals	2.0 Hrs
3	Session 3	**Understanding view manipulation** Fly mode Fit all In Pan Rotate Zoom in Zoom out Normal view Multi view Quick view View mode Hide/Show Swap visible space **Understanding standard toolbar** Save management Print Cut Copy Paste Undo Redo **Understanding sketch tools** **Creating construction/ standard element** **Applying geometrical/ dimensional constraints** **Understanding user selection filter toolbar** **Understanding visualization toolbar** Cut part by sketch plane 3D Element visualization D Element visualization Diagnostics Dimensional constraints	Theory/Demo	2.0 Hrs

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		Geometrical constraints		
		Sketch Workflow		
4	Session 4	Exercises included for practice	Practicals	2.0 Hrs
5	Session 5	**Editing profile**	Theory/Demo	2.0 Hrs
		Corner		
		Chamfer		
		Relimitations		
		Transformation		
		3D geometry		
		Understanding constraints		
		Constraints		
		Constraint creation		
		Contact constraint		
		Fix together		
		Auto constraint		
		Animate constraint		
		Edit Multi – constraint		
		Introduction to part design		
		Creating sketch-based features		
		Pad definition		
		Drafted filleted pad definition		
		Pocket definition		
		Drafted filleted pocket		
		Shaft definition		
		Groove definition		
		Hole definition		
		Creating reference element		
		Plane definition		
6	Session 6	Exercises included for practice	Practicals	2.0 Hrs
7	Session 7	**Creating sketch-based feature**	Theory/Demo	2.0 Hrs
		Rib definition		
		Slot definition		
		Stiffener definition		
		Solid combine definition		
		Applying dress-up features		
		Edge fillet definition		
		Variable fillet definition		
		Chordal fillet definition		
		Face-face fillet definition		
		Tritangent fillet definition		
		Chamfer definition		
8	Session 8	Exercises included for practice	Practicals	2.0 Hrs
9	Session 9	**Creating sketch-based features**	Theory/Demo	2.0 Hrs
		Multi-sections solid definition		
		Removed multi-section solid definition		
		Applying dress-up features		
		Draft definition		
		Draft reflect line definition		
		Variable angle draft definition		
		Shell definition		
		Thickness definition		
		Thread/tap definition		
		Remove/replace face definition		
		Creating Surface-Based Features		
		Split Definition		
		Thick Surface Definition		
		Close Surface Definition		
		Sew Surface Definition		
10	Session 10	Exercises included for practice	Practicals	2.0 Hrs
11	Session 11	**Creating reference element**	Theory/Demo	2.0 Hrs
		Point definition		
		Line definition		
		Creating transformation features		
		Transformation definition		
		Rotation definition		

		Symmetry definition		
		Axis to Axis definition		
		Mirror Definition		
		Pattern – Rectangular/ Circular pattern definition		
		Scale definition		
12	Session 12	Exercises included for practice	Practicals	2.0 Hrs
13	Session 13	**Introduction to Assembly Design**	Theory/Demo	2.0 Hrs
		Understanding top down & bottom up approach		
		Understanding Product Structure Tools		
		New component		
		New product		
		New part		
		Existing component		
		Existing component with positioning		
		Understanding Constraints Toolbar		
		Coincidence constraints		
		Contact constraints		
		Offset constraints		
		Angle constraints		
		Fix component		
		Fix together		
		Quick constraint		
		Flexible/Rigid sub-assembly		
		Change constraint		
		Reuse pattern		
14	Session 14	Exercises included for practice	Practicals	2.0 Hrs
15	Session 15	**Introduction to sheet metal design**	Theory/Demo	2.0 Hrs
		Understanding sheet metal parameters		
		Understanding views toolbar		
		Fold/Unfold		
		Multi viewer		
		Views management		
		Creating walls		
		Wall		
		Wall on edge		
		Extrusion		
		Swept walls- Flange, Hem, Tear Drop, User Flange		
		Creating rolled walls		
		Hopper		
		Free from surface		
		Rolled wall		
		Understanding bending operations		
		Bend / Conical bend		
		Bend from flat		
		Unfolding/ Folding		
		Point or curve mapping		
		Recognize definition		
16	Session 16	Exercises included for practice	Practicals	2.0 Hrs
17	Session 17	**Wireframe and Surface Design**	Theory/Demo	2.0 Hrs
		Introduction to Surface Design		
		Understanding Wireframe Toolbar		
		Point, point and planes repetition		
		Line, axis, polyline		
		Plane, planes between		
		Projection		
		Intersection		
		Circle		
		Corner		
		Connect		
		Spline		
		Helix		
		Creating Surfaces		
		Extruded surface definition		
		Revolution surface definition		
		Sphere surface definition		

		Trim & Split Definition		
		Creating Surfaces		
		Cylinder surface definition		
		Offset surface definition		
		Fill surface definition		
		Multi-Sections surface definition		
		Blend definition		
		Creating Sweep Surface		
		Swept surface definition using profile type: -		
		Explicit		
		Line		
		Circle		
		Conic		
18	Session 18	Exercises included for practice	Practicals	2.0 Hrs
19	Session 19	**Introduction to drafting**	Theory/Demo	2.0 Hrs
		Creating new drawing file		
		Creating projection views		
		Front view		
		Un folded view		
		View from 3D		
		Projection		
		Auxiliary		
		Isometric		
		Advanced front view		
		Creating sectional views		
		Offset section view		
		Aligned section view		
		Offset section cut		
		Aligned section cut		
		Creating details views		
		Detail		
		Sketched detail profile		
		Quick detail		
		Sketched quick detail profile		
		Creating Dimensions Using		
		Dimensions		
		Chained Dimensions		
		Cumulated Dimensions		
		Stacked Dimensions		
		Length Dimension		
		Angle Dimension		
		Radius Dimension		
		Diameter Dimension		
20	Session 20	Exercises included for practice	Practicals	2.0 Hrs